ASHUTOSH SINGH

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CAREER SUMMARY:

Full Stack Developer with 3+ years of experience delivering scalable web and enterprise solutions. Currently working at Tata Technologies in Apriso (MES), contributing to both application development and production support. Experienced in Apriso, React, Next.js, Django, Flask, MSSQL, and REST APIs. Strong expertise in performance optimization, manufacturing systems, and building business-driven applications.

WORK EXPERIENCE:

Solution Developer | Tata Technologies
Apr 2024 – Present

- Developed and supported enterprise-grade manufacturing solutions using **Apriso (MES)** and aligned with business workflow requirements.
- Designed and implemented **workflow-driven applications** for production operations, issue tracking, approval processes, and shop-floor automation.
- Built 1**0+ FTPC and Java-based MES applications** for the CORAL manufacturing line with role-based workflows and process validations.
- Built custom dashboards and reporting applications for production monitoring, defect tracking, and operational insights.
- Developed **real-time monitoring dashboards with voice alerts** for critical production events and system failures.
- Designed and developed an **end-to-end BIW shop web application** with separate modules for order start, booking, punching, defect logging, and workflow tracking.
- Automated BOM validation and notification workflows using **SQL Server procedures and email-based alert systems.**
- Optimized complex SQL queries and stored procedures to improve system performance and reduce execution time.
- Maintained detailed technical documentation for workflows, integrations, and system configurations.

Full Stack Developer | TechPath Research & Development
Jul 2023 – Feb 2024

- Developed scalable backend systems using **Django and Django REST Framework** for enterprise and client-facing applications.
- Built a **complete e-commerce backend architecture** with product variations, cart, orders, authentication, and payment integration.
- Developed a **Project Management System** with authentication, role-based access, workflow tracking, and task assignment modules..
- Designed REST APIs for scalable communication between frontend and backend systems.
- Developed multiple React-based applications with real-time API integrations and state management.
- Integrated Razorpay payment gateway for secure checkout and order processing workflows.

Python Intern | TechPath Research & Development
Dec 2022 – Jun 2023

- Developed Flask-based web applications with complete CRUD functionality and authentication workflows.
- Built a **blog management application** with user registration, login, post creation, editing, and deletion modules.
- Developed backend logic for session management, route protection, and secure data handling.
- Implemented SQLAlchemy-based database models and optimized database operations.
- Developed a **Gym Management System frontend** with secure payment workflow integration using Razorpay.
- Gained hands-on experience in backend architecture, database design, and enterprise coding standards.

EDUCATION:

Master's of Computer Application (MCA)
Manipal University

09/2023 - 11/2025 **Jaipur, India**

Bachelor's of Computer Application (BCA)
Mahatma Gandhi Kashi Vidyapeeth

11/2020 - 09/2023 **Varanasi, India**

SKILLS:

- **Backend & Programming:** Python, Django, Django REST Framework, Flask, JavaScript, Java
- **Frontend & UI:** React.js, Next.js, React Native, Tailwind CSS, Bootstrap, Framer Motion
- **Database & Tools:** MSSQL Server, SQL, GIT, SQLAlchemy
- **Enterprise & Other:** Apriso (MES), FTPC, Angular (Basics), SCSS, DSA, C, C++

PROJECTS:

BIW Shop MES Application
Tech Stack: React.js, Node.js, Express.js, MSSQL

- Designed and developed a **full-scale Manufacturing Execution System (MES)** for BIW shop, digitizing and streamlining end-to-end shop-floor operations.
- Built a modular and scalable architecture using **React.js (frontend)** and **Node.js + Express.js (backend)** with **MSSQL** for high-performance data handling.
- Developed and implemented key production modules such as **Order Start, Booking, Punching, Defect Logging** etc.
- Enabled **real-time data flow and synchronization** between shop-floor systems and backend servers to ensure accurate production tracking.
- Designed **RESTful APIs** for seamless communication between frontend interfaces and backend services.

CORAL MES Applications (Rockwell FTPC)
Tech Stack: Rockwell FTPC, Java-based MES Tools

- Developed and deployed multiple **Windows-based Manufacturing Execution System (MES) applications** for the CORAL production line using **Rockwell FTPC** and Java-based tools.
- Built critical shop-floor applications covering complete production lifecycle, including **Order Start, Booking, Dressing, PDI, Leak Test, Load Test** etc.
- Designed and implemented **role-based workflows**, ensuring secure and controlled access for operators, supervisors, and administrators.
- Ensured **seamless interaction between MES applications and shop-floor systems**, enabling real-time execution and monitoring of manufacturing processes.
- Provided **production support and enhancements**, resolving live issues and improving system performance.

Vantage Reporting System
Tech Stack: React.js, Node.js, Express.js, MSSQL

- Designed and developed an i**nteractive web-based reporting system** for real time MES data.
- Built a scalable architecture using **React.js for frontend** and **Node.js + Express.js for backend**, with **MSSQL** handling large-scale production data.
- Developed **dynamic dashboards** with rich data visualizations (charts, tables, and summary views) and integrated **interactive UI features and animations** to enhance user experience and improve data readability for better decision-making.
- Built functionality for **downloading reports** in structured formats, supporting offline analysis and reporting needs.
- Designed and optimized backend APIs for **fast data retrieval and efficient handling of complex queries.**

Real-Time Monitoring Portal
Tech Stack: React.js, KEPServer APIs

- Developed a **real-time monitoring dashboard** to track live production data and system events across manufacturing lines and display **real-time metrics, alerts, and system statuses** for quick analysis.
- Integrated **KEPServer APIs** to stream live data from shop-floor machines and industrial systems.
- Implemented an intelligent **voice alert system** that automatically notifies users of critical production issues, including **Line stoppages, High event counts, Live Trasfer failures** etc.